K BALAJI

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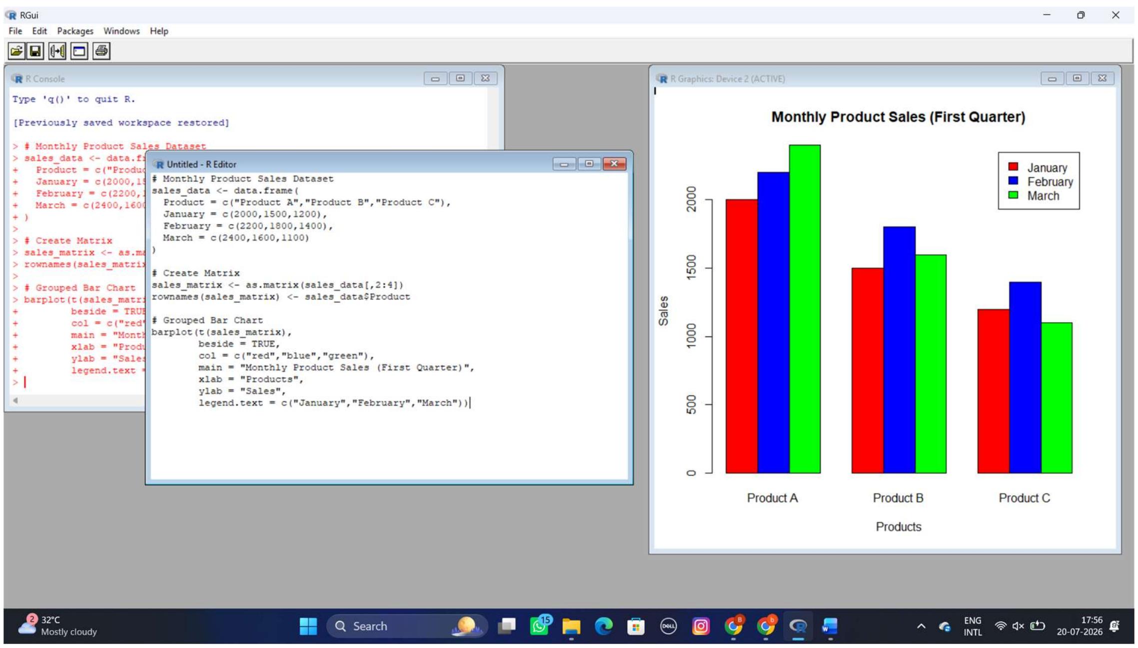
DATA HANDLING AND VISUVALIZATION LAB PROGRAMS 6-10:

EXPERIMENT 6:

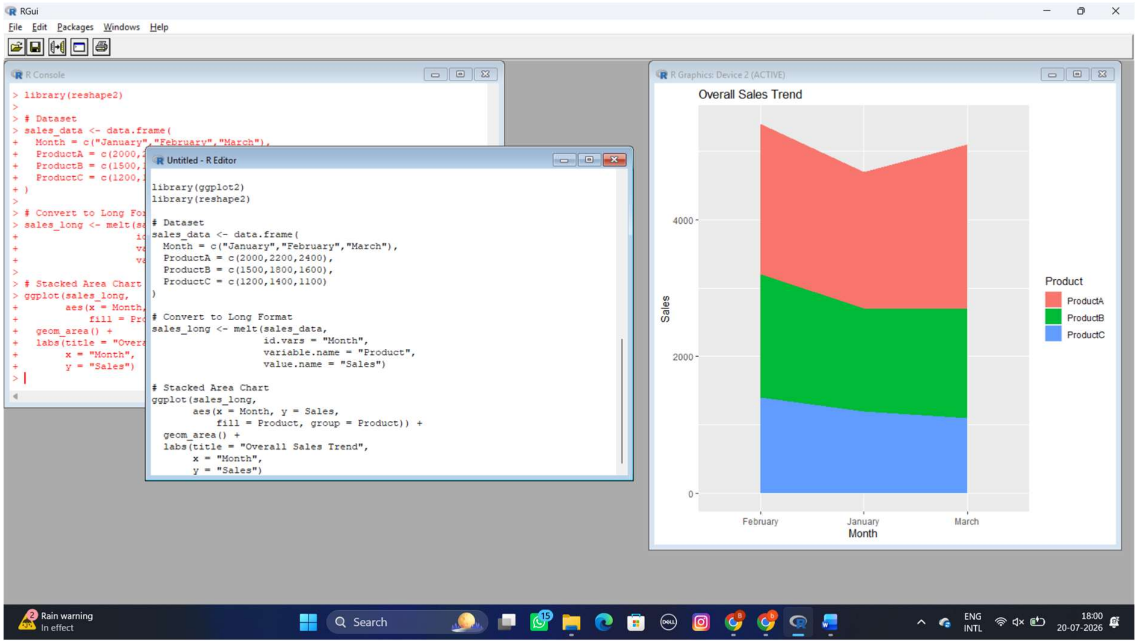
Product Sales Analysis

Dataset: Monthly Product Sales

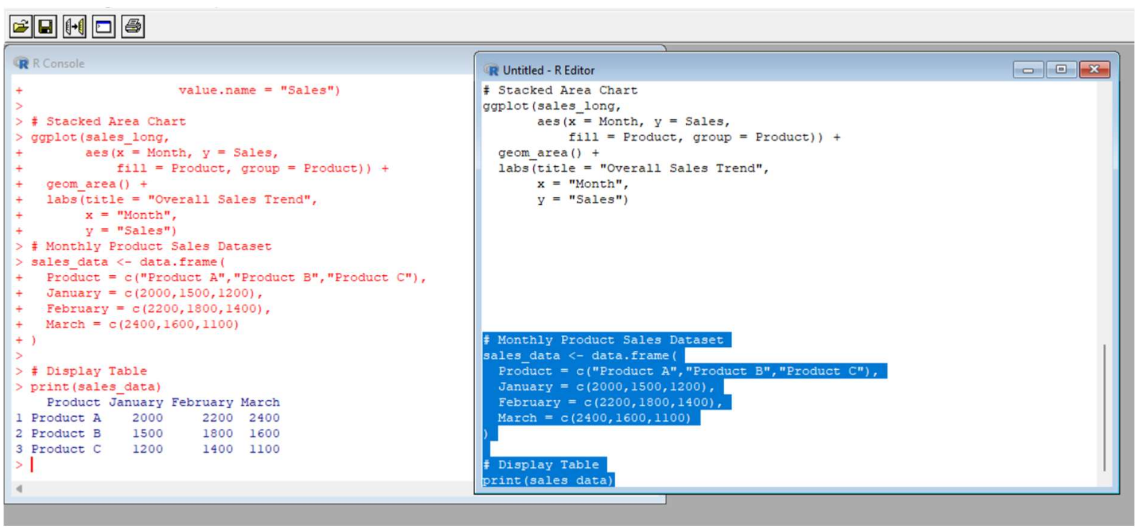
1)Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.



2) Generate a stacked area chart to represent the overall sales trend for all products over the three months.



3) Build a table to show the monthly sales data for each product. Label the table elements.

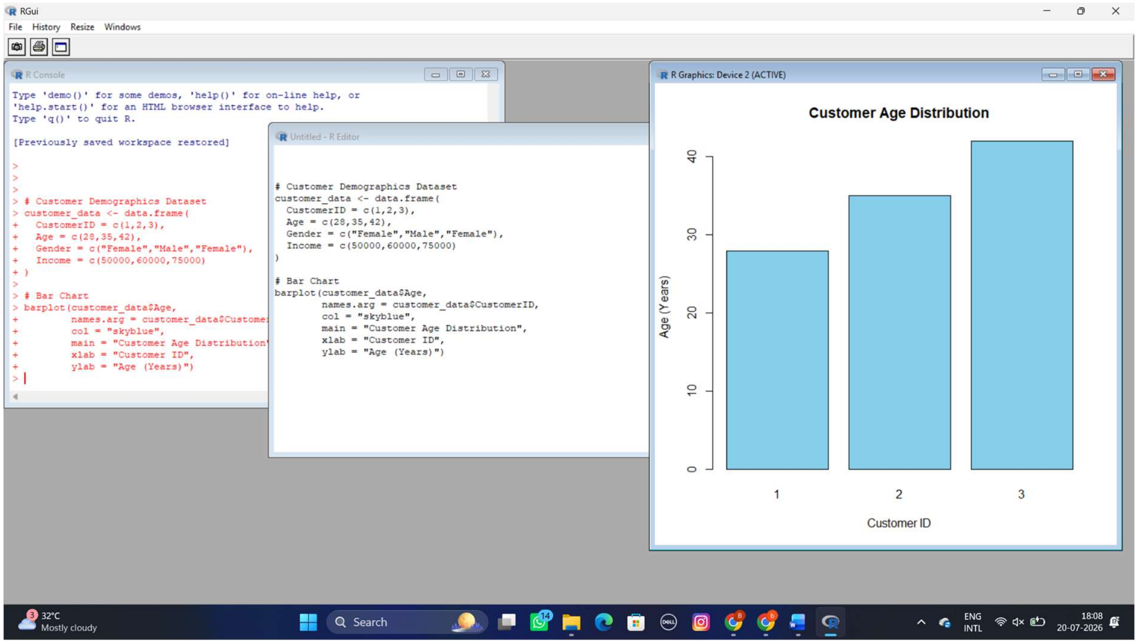


EXPERIMENT 7:

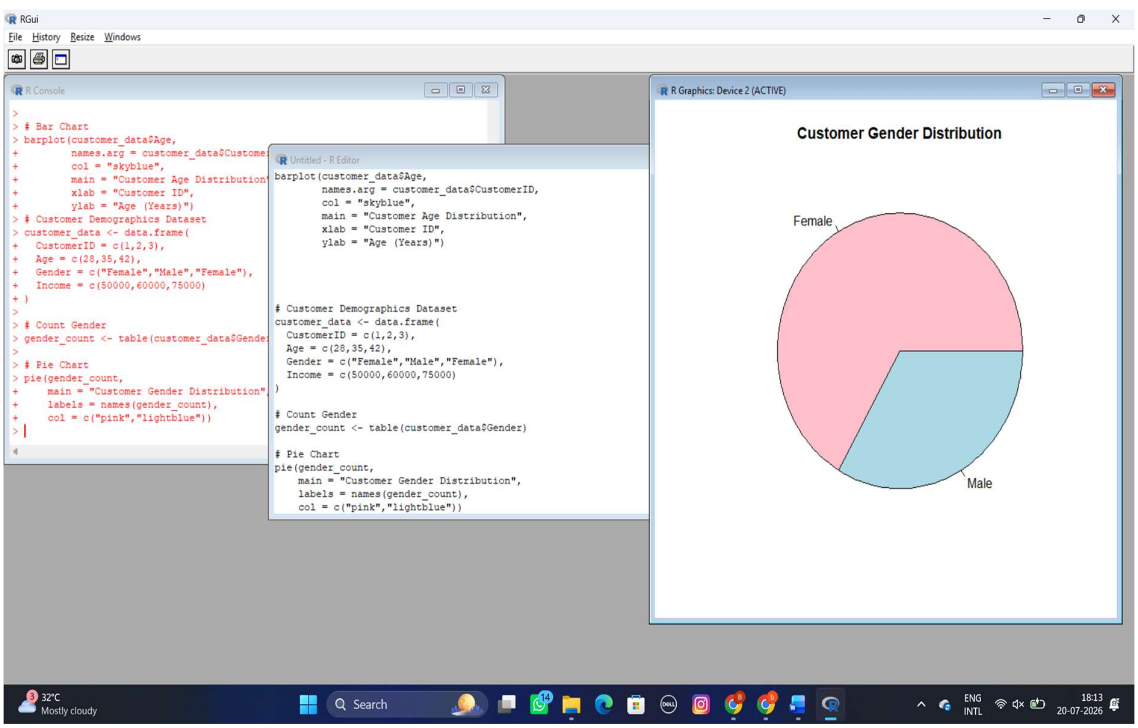
Customer Demographics Analysis

Dataset: Customer Demographics

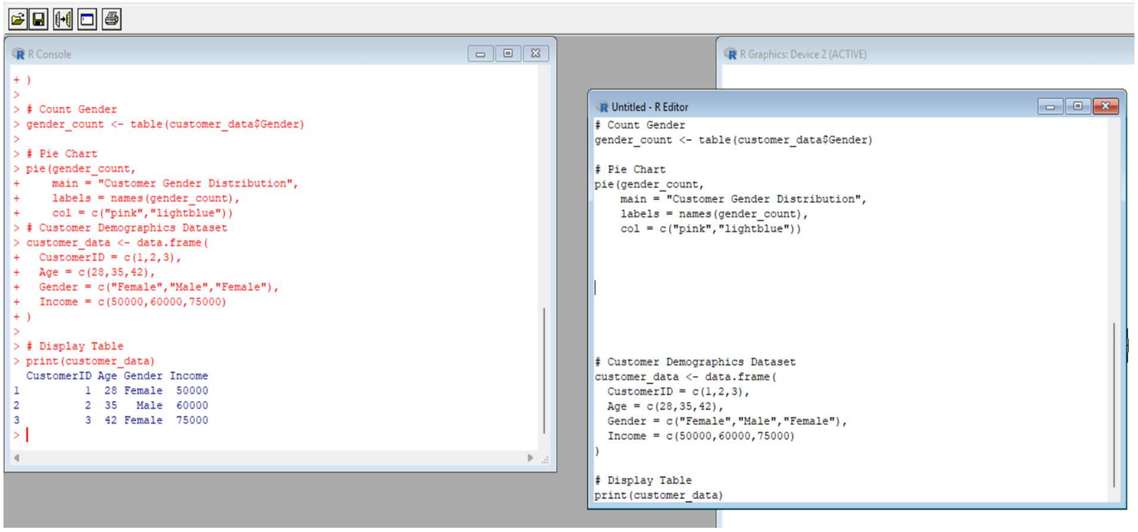
1) Create a bar chart to visualize the distribution of customer ages. Label the axes and title the chart.



2) Generate a pie chart to represent the distribution of customers by gender.



3) Build a table to show the demographic information for each customer. Label the table elements.

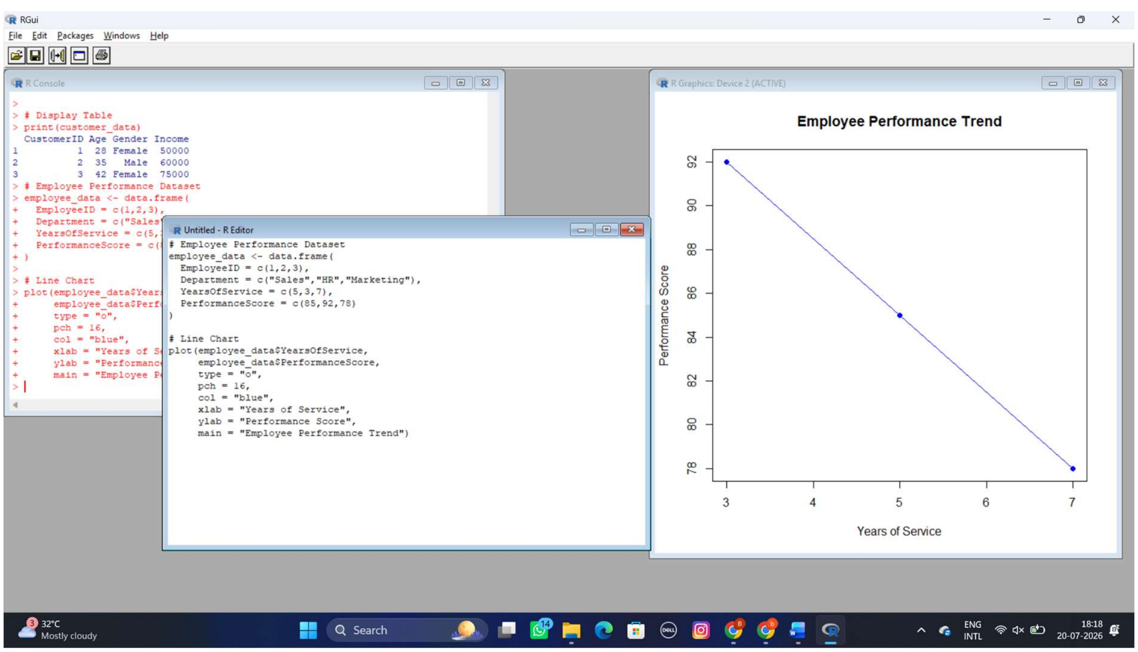


EXPERIMENT 8:

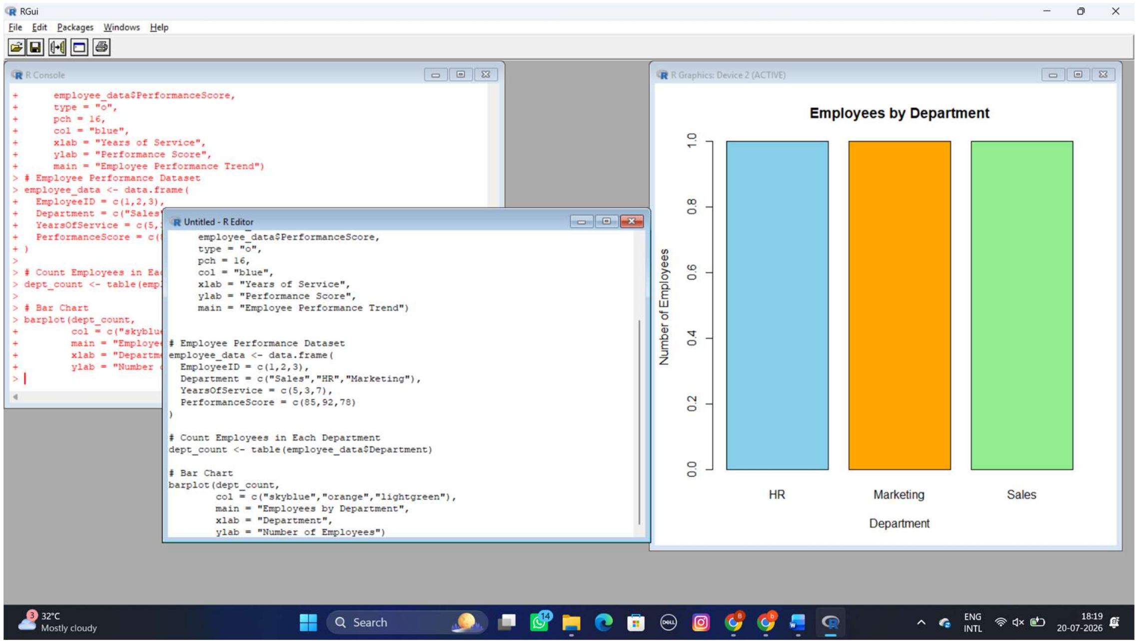
Employee Performance Analysis

Dataset: Employee Performance

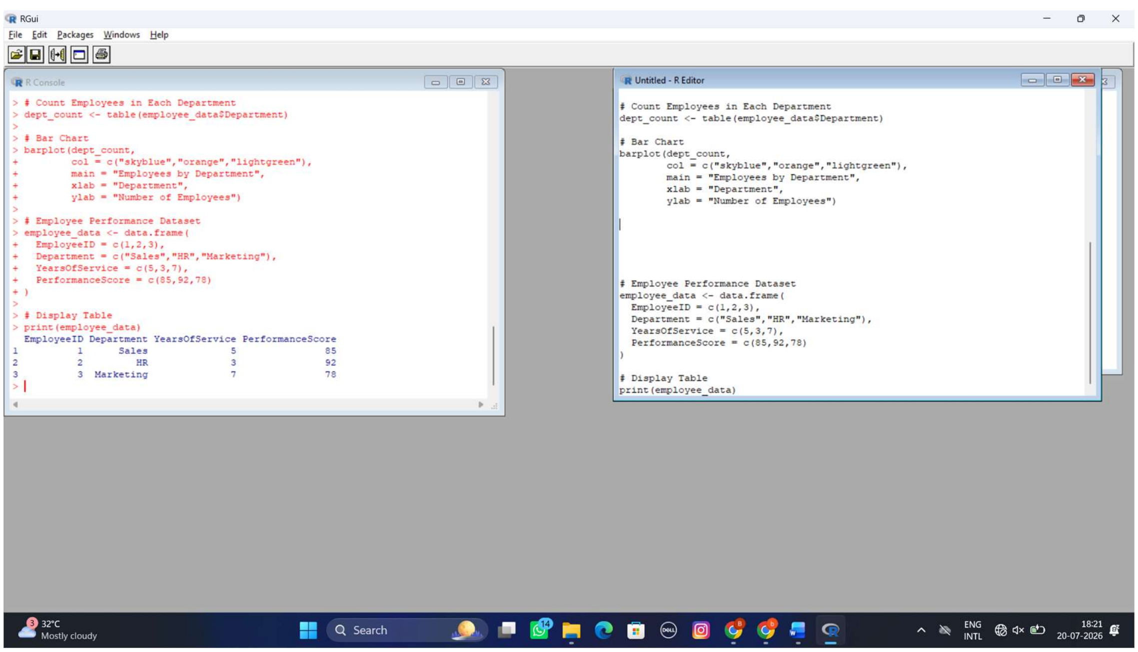
1) Create a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.



2) Generate a bar chart showing the distribution of employees across different departments. Label the chart elements.



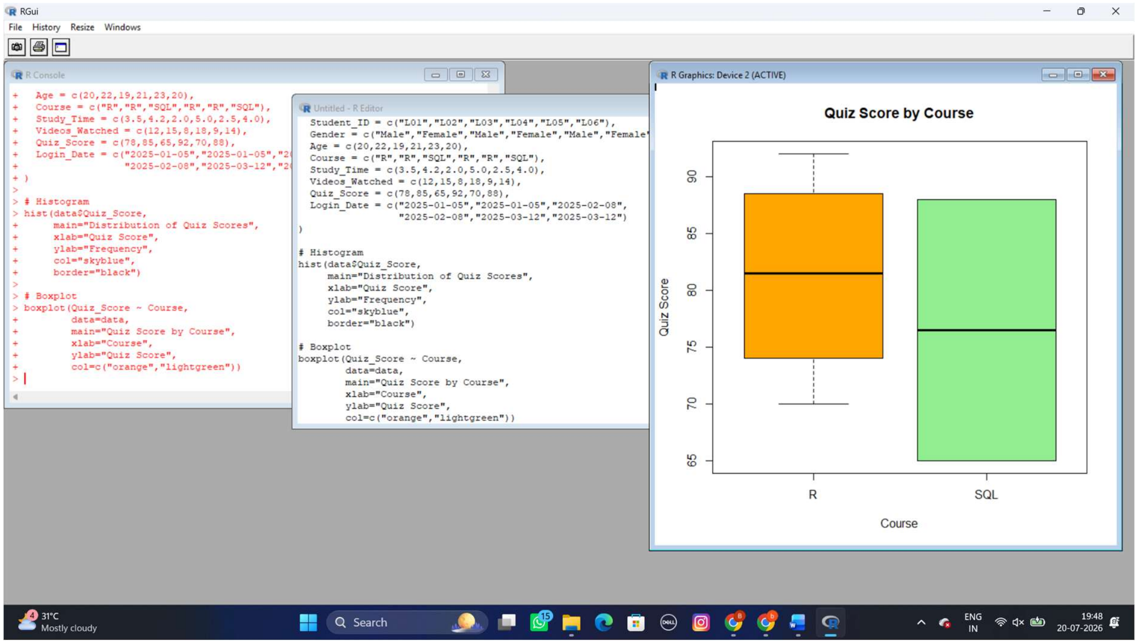
3) Build a table to display the performance data for each employee. Label the table elements.



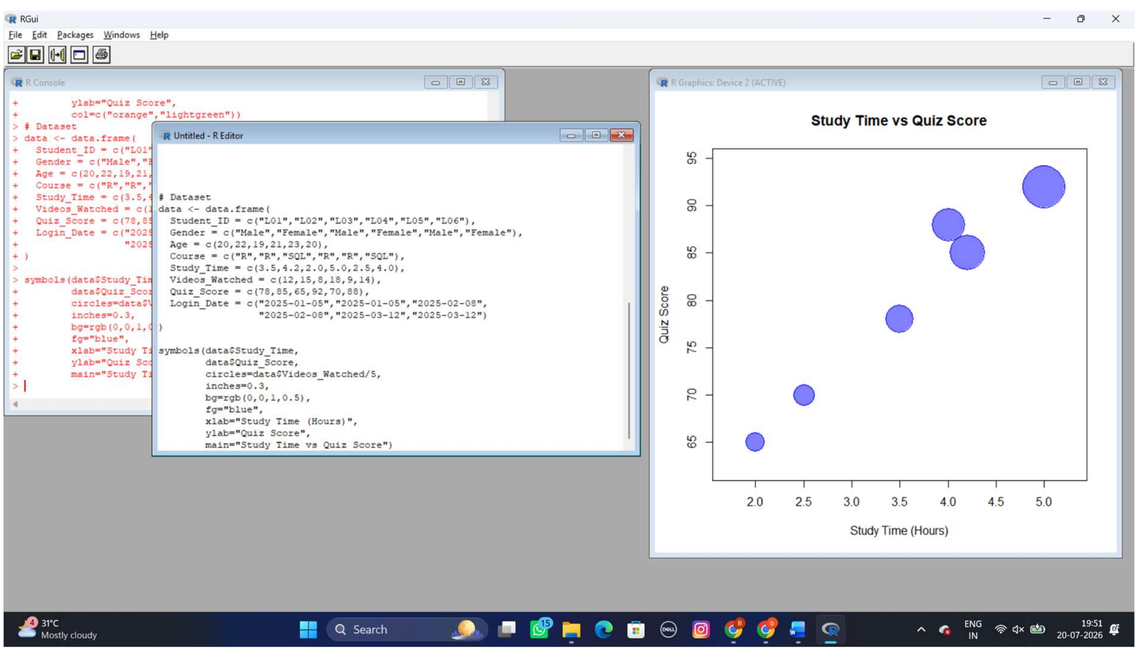
EXPERIMENT 9:

Dataset: Online_Learning_Activity.csv

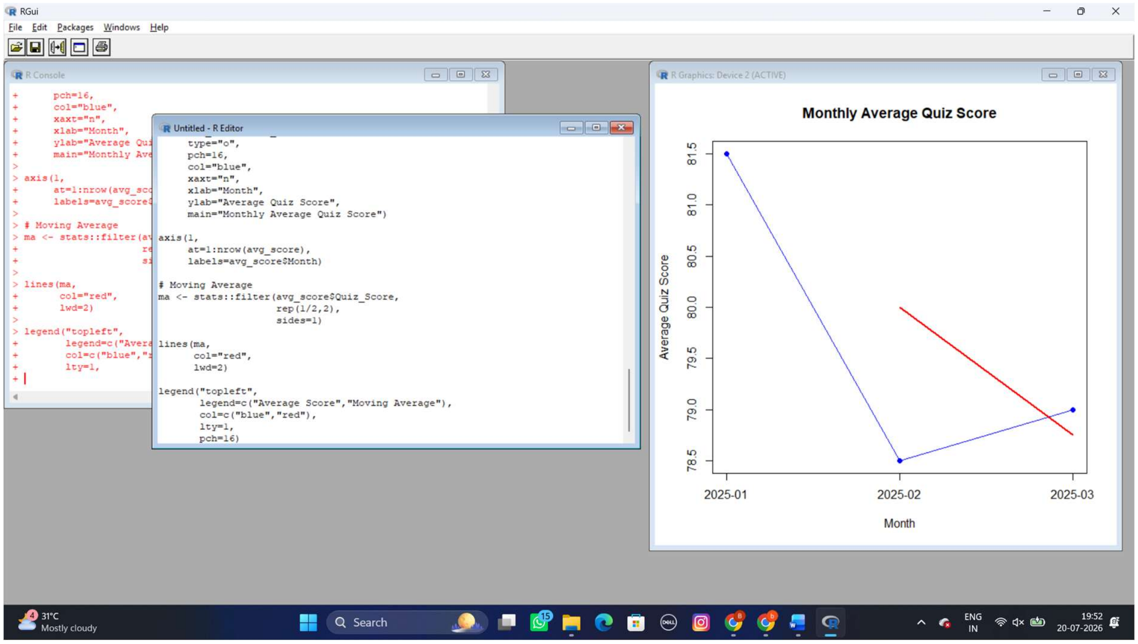
1) Import the dataset in R. Plot a histogram for Quiz_Score. Draw a boxplot of Quiz_Score by Course. Add title and labels. Interpret student performance.



2) Create a scatter plot using R programming between Study_Time and Quiz_Score. Represent Videos_Watched using bubble size. Apply transparency. Explain the learning pattern.



3) Using R programming, convert Login_Date into Date format. Calculate average quiz score per month. Plot a line chart. Apply moving average smoothing. Identify the trend.

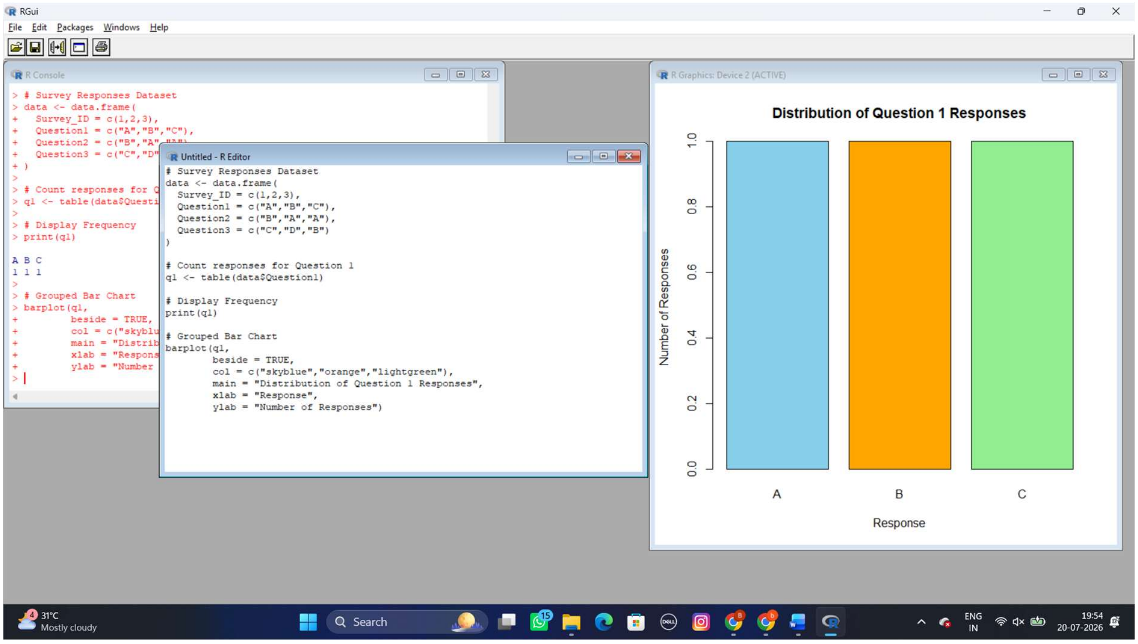


EXPERIMENT 10:

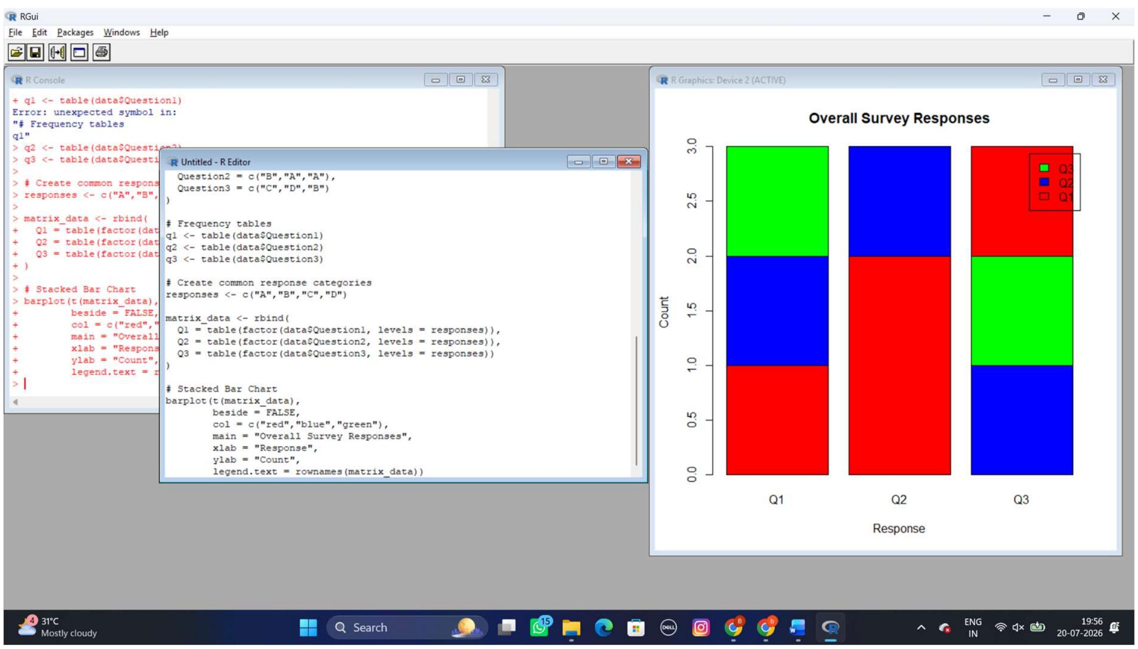
Survey Responses Analysis

Dataset: Survey Responses

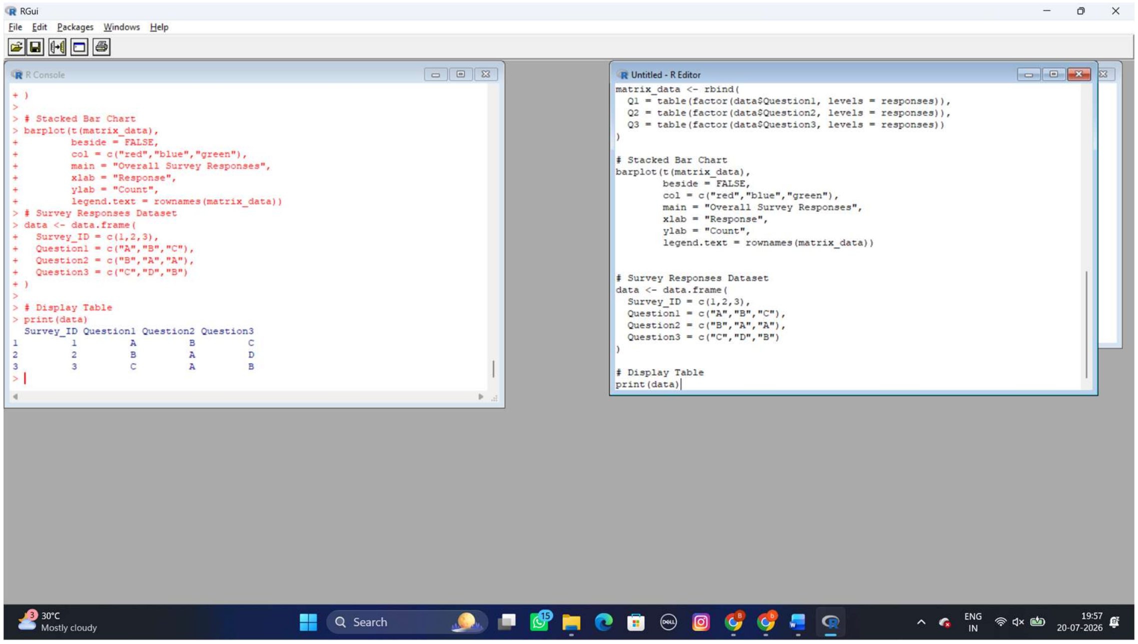
1) 1.Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.



2) Build a table to display the sales data for each product category. Label the table elements.



3) Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.


RGui
File Edit Packages Windows Help
R Console
+)
>
> # Stacked Bar Chart
> barplot(t(matrix_data),
+ beside = FALSE,
+ col = c("red","blue","green"),
+ main = "Overall Survey Responses",
+ xlab = "Response",
+ ylab = "Count",
+ legend.text = rownames(matrix_data))
> # Survey Responses Dataset
> data <- data.frame(
+ Survey_ID = c(1,2,3),
+ Question1 = c("A","B","C"),
+ Question2 = c("B","A","A"),
+ Question3 = c("C","D","B")
+)
>
> # Display Table
> print(data)
Survey_ID Question1 Question2 Question3
1 1 A B C
2 2 B A D
3 3 C A B
>
Untitled - R Editor
matrix_data <- rbind(
Q1 = table(factor(data$Question1, levels = responses)),
Q2 = table(factor(data$Question2, levels = responses)),
Q3 = table(factor(data$Question3, levels = responses))
)
Stacked Bar Chart
barplot(t(matrix_data),
beside = FALSE,
col = c("red","blue","green"),
main = "Overall Survey Responses",
xlab = "Response",
ylab = "Count",
legend.text = rownames(matrix_data))
# Survey Responses Dataset
data <- data.frame(
Survey_ID = c(1,2,3),
Question1 = c("A","B","C"),
Question2 = c("B","A","A"),
Question3 = c("C","D","B")
)
# Display Table
print(data)
30°C
Mostly cloudy
Search
ENG
IN
19:57
20-07-2026